

KRISHNA JAMAKHANDI

Bioengineering @ UC San Diego

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SKILLS

- Java/Python
- MATLAB
- Fusion 360/Onshape
- Arduino
- Microsoft Office
- Laboratory Equipment Operation and Maintenance
- Electrophoresis Operation
- Chromatography Operation
- Calorimetry Operation
- Buffer Preparation
- Chemical Solution Creation
- Data Analysis
- Critical Thinking
- Sense of urgency, drive, ownership
- Cross-functional collaboration

VOLUNTEERING

Scout Eagle Project

Creator and Architect

Feb 2024 - May 2024

- Created a schematic for five benches for the local community temple
- Organized lists of materials and managed costs for the entire project
- Was the key organizer with the non-profit for which the benches were created
- Led 20+ scouts through the creation and painting of the five benches

Eagle Scout Leader

Boy Scouts of America

Jan 2020 - Dec 2024

- Built the logic for a troop website to inform parents of happenings and important documents regarding their scouts

CAREER OBJECTIVE

I am a second-year student at UC San Diego with an avid interest in bioengineering, biotechnology, and their intersection in the robotic world. With a strong academic foundation and my experience in software, hardware and electronics, I'm eager to apply my skills and knowledge to explore the cutting edge of science and technology in the medical field. Parallel to my studies, I have honed leadership, teamwork, and problem-solving abilities, while maintaining a commitment to community service and personal development. I am enthusiastic about using biological systems to address real-world health challenges and am excited to connect with professionals and peers who share similar interests in advancing science and technology.

EDUCATION

Second Year, Bioengineering

UC San Diego

📅 September 2024 - current

Relevant courses

- Engaged with cardiac electrophysiology and skeletal muscle electrophysiology by analyzing heart rate, pulse rate, and PQRST waves to study electrical activity and physiological responses.
- Gained understanding in spinal anatomy by studying the five regions of the human spine, and analyzed vertebral structures, intervertebral discs, and their roles in structural support and spinal cord protection.
- Simulated bioelectrical conductivity and impedance in tissues through the use of electrodes in saline solutions, calculating voltage and current using Ohm's Law
- Studied the biomechanics around scoliosis, experimenting on simulations of its structural vs material properties
- Studied the inner workings of the knee, specifically focusing on the ACL and MCL and their roles in the biomechanics of the knee joint

EXPERIENCE

Undergraduate Student Researcher, DVJ Lab

UC San Diego Jacobs School of Engineering

📅 Sep 2025 - current 📍 La Jolla, CA

- Contributing to research on cardiovascular adaptations to pulmonary arterial hypertension (PAH), studied the role of the renin-angiotensin-aldosterone system (RAAS) in disease progression.
- Assisting in rat surgeries by preparing instruments, collecting tissues, and weighing organs for post-operative analysis.
- Performing wet lab techniques such as membrane preparation, RNA isolation, and tissue staining under the mentorship of a PhD student.
- Supporting ongoing experiments by helping collect and organize data from animal models, contributing to translational research that connects cellular activity to systemic function
- Conducting Planar Biaxing of right ventricular heart tissue to compare the effect of the progression of PAH in the extracellular matrix in live and decellularized samples

Electrical/Software Sublead

Biomedical Engineering Society @ University of California, San Diego

📅 Oct 2025 - current 📍 La Jolla, CA

- Leading software and electrical integration for a prosthetic arm and haptic glove system, coordinating signal flow between position-tracking sensors, actuators, and embedded controllers across multiple subassemblies.

- Organized and maintained camping and medical supplies and legal/safety contracts
- Oversaw 20+ scouts in activities and overnight campouts
- Led 15+ scouts in volunteering activities for non-profit organizations

Club Co-President

Chemistry Club @ Mission San Jose High School

Sep 2023 - May 2024

- Organized events to engage people and encourage their participation in meetings, sparking their interest in the club
- Taught students AP Chemistry curriculum through cram sessions and weekly meetings
- Created easy to understand presentations to prepare students for USNCO exams

Student Teacher

West Coast World Martial Arts

Dec 2023 - May 2024

- Instructed children in the fundamentals of martial arts, with a focus on Taekwondo techniques.

Assistant Teacher at Language School

Kannada Kali

Sep 2020 - May 2021

- Taught children the basics of language fundamentals, focusing on vocabulary and communication skills and assisted teacher with classroom tasks

- Designing and debugging sensor-actuator pipelines, translating glove position tracking data into linear actuator outputs for prosthetic motion control.
- Developing and evaluating embedded hardware architectures (Arduino-based), addressing I/O constraints for multi-actuator systems through I2C communication, multiplexers, and driver ICs.
- Implementing and testing haptic feedback systems using vibrotactile motors with motor drivers, enabling graded vibration intensity via PWM control.
- Investigating and learning PCB design and fabrication workflows using KiCad (prototyping) and Altium (manufacturing)
- Integrating IMUs, encoders, and analog sensors for wrist and thumb tracking; exploring DOF tradeoffs for finger and wrist kinematics.
- Collaborating cross-functionally with mechanical and software subteams, participating in subsystem reviews, signal architecture discussions, and rapid design iteration.

Programming & Hardware Lead

First Tech Challenge

📅 Sep 2022 - May 2024 📍 Fremont, CA

- Co-created a library for OpenCV for use during competition
- Built the logic for autonomous and teleoperation sections
- Mentored younger members by teaching Java and encouraging code logic improvement
- Co-created and taught in Java classes for children under the age of fifteen
- Led team to two consecutive regional berths and achieved awards during the regional competitions
- Self-taught CAD designing and 3D printing software
- Self-taught and used CNC machining to create custom parts from original designs
- Taught and built CAD designs for custom implements to the robot

Chemical Technician

Mission San Jose High School

📅 Aug 2023 - May 2024 📍 Fremont, CA

- Handled laboratory equipment inventory and cleaning
- Spent 40+ hours testing and verifying the quality of spectrophotometers and calorimeters
- Performed 15+ experiments before laboratory time to gather data models for students